Pendulum swing of brass nut for gravity measurement

Run r_20261101_042427_ca6b6d · Date 2026-11-01 · Verdict: PASS WITH WARNINGS

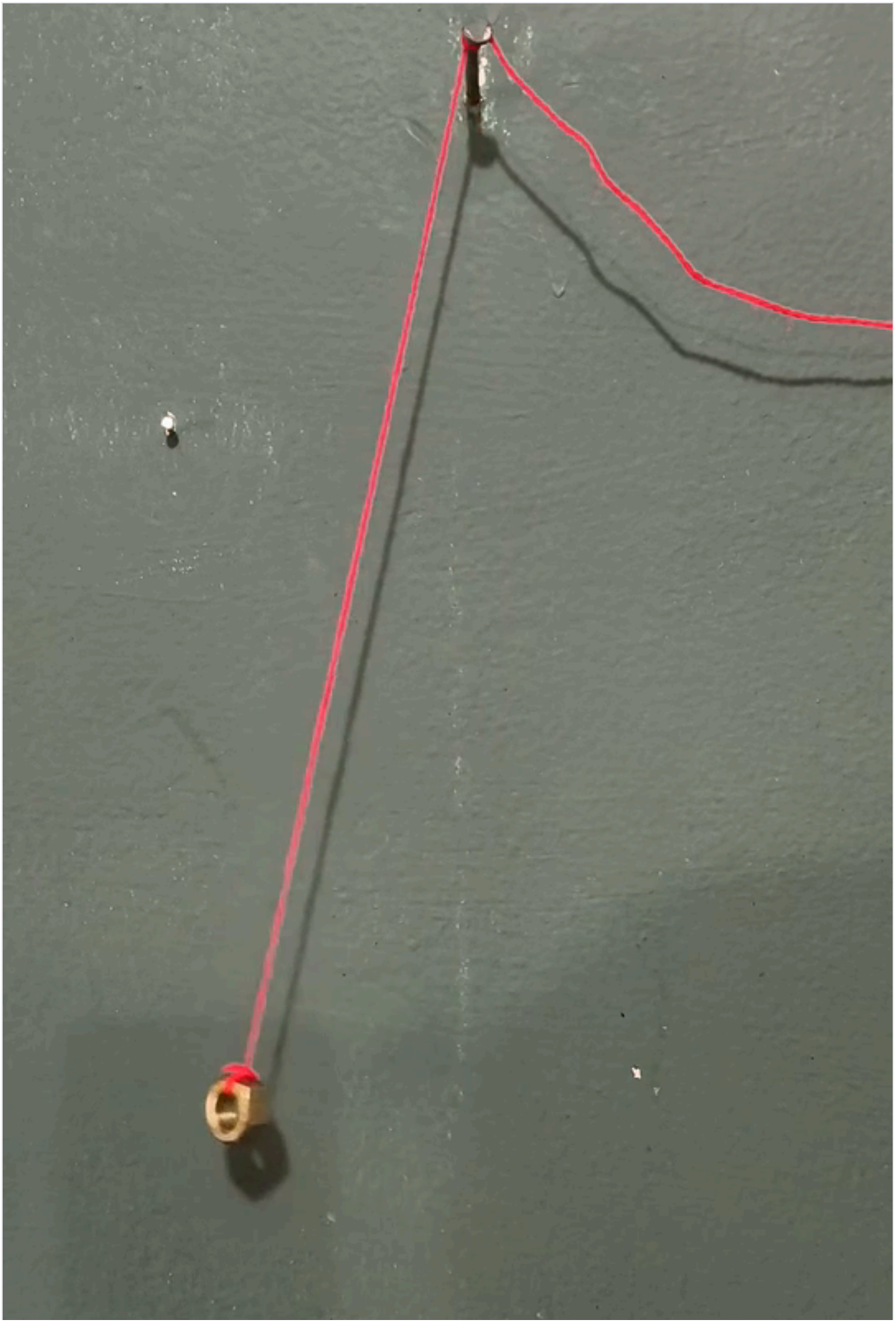
$g = 9.62 \pm 0.17\ m/s^2$

Aim

To measure gravity from the swing of a hanging brass nut and to test the predicted period.

Setup

A brass nut hangs from a red string pinned to a wall and swings as a pendulum. A hand briefly enters during release and some blur is visible during fast motion.





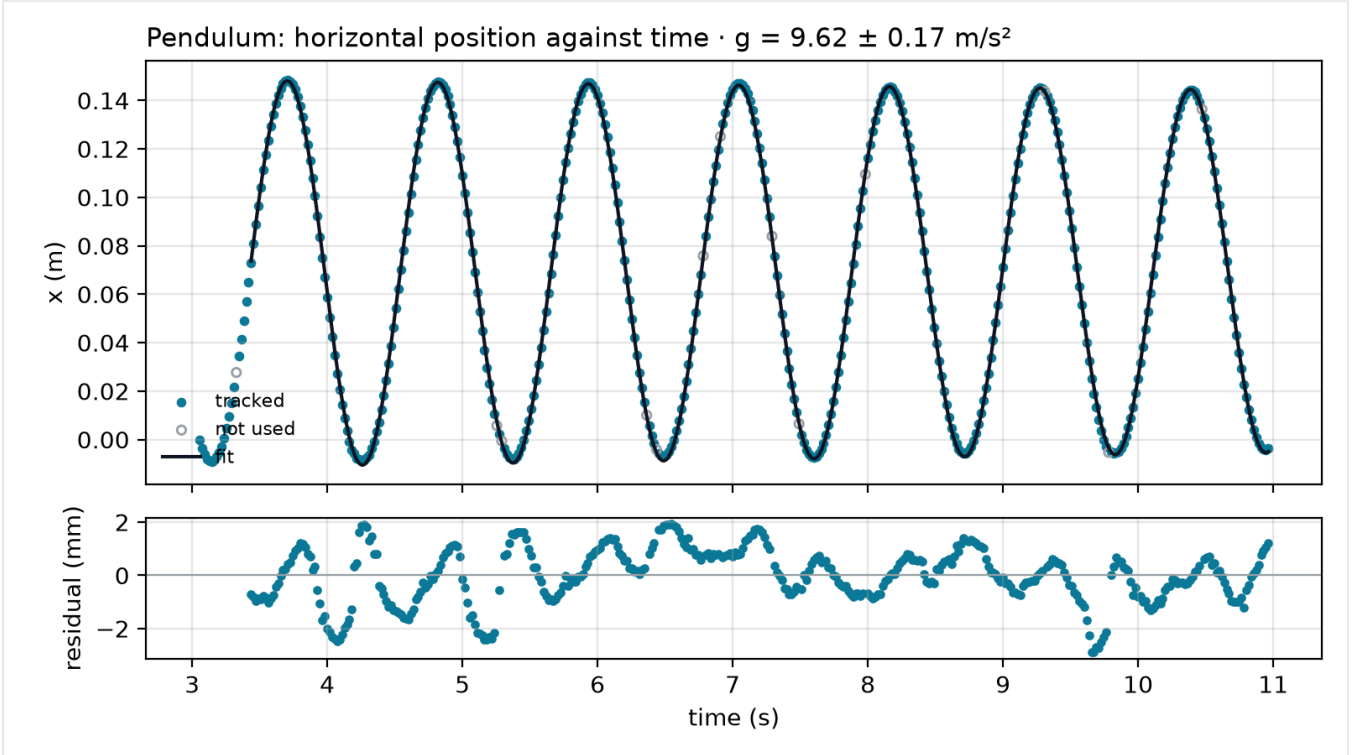
Key frame

Method

We tracked the nut frame by frame, trimmed the release to keep clean swinging, fitted a damped sine curve to the sideways motion, estimated uncertainty by resampling, corrected the period for swing size, and converted the period to gravity using the stated string length. Fit quality is The damped sine fit explains the motion very well with low residuals. and the scale from the spoken length is scale = 0.461 mm/px.

Model	Role in this run
Muse Voice Transcribe	speech → transcript
Muse Spark 1.3	Planner, Analyst, Auditor, Translator
SAM 3.1	object tracking on the video and the ruler
Muse Wonder 1.0	not used in this run
Muse Spark Open (open weights, via Baseten)	Designer

Results



Trajectory

Gravity and period

The headline gravity is g = 9.62 ± 0.17 m/s² with reference comparison a difference of -1.62% from the reference value. The measured period is period = 1.11412 ± 0.00024 s and the size corrected value is period0 = 1.1094 ± 0.0024 s. The resampling interval is period = 1.1094 ± 0.0024 s.

Calibration and quality

Scale and fit details are scale = 0.461 mm/px and The damped sine fit explains the motion very well with low residuals..

Hypothesis evaluation

The predicted period is not supported, since the measured period period = 1.11412 ± 0.00024 s differs clearly and the formal test gives not supported.

Error analysis

Main uncertainties come from the stated string length, pivot and scale estimation, motion blur during fast swings, brief hand contact at release, and small wobble not captured by a simple damped sine. The resampling interval period = 1.1094 ± 0.0024 s captures timing variation but not systematic length bias.

Conclusion

The pendulum gives a gravity value g = 9.62 ± 0.17 m/s² consistent with the reference within a difference of -1.62% from the reference value, supported by a strong fit The damped sine fit explains the motion very well with low residuals..

Next experiment

Pendulum Period vs String Length / Periode Pendulum vs Panjang Tali

Based on pendulum law, increasing length increases period. Predicted periods are 1.43 s for the shortest length, 2.03 s for the next, 2.48 s for the next, and 2.86 s for the longest length. A plot of period squared versus length should be a straight line through the origin.

- 0.500 m → 1.43 s
- 1.00 m → 2.03 s
- 1.50 m → 2.48 s
- 2.00 m → 2.86 s

Controlled variables: Bob mass and shape kept identical / Massa dan bentuk bandul dijaga identik, Release angle kept small and identical / Sudut pelepasan dijaga kecil dan identik, Same support point and indoor location, no wind / Titik tumpu dan lokasi dalam ruangan sama, tanpa angin, Same timing method - phone slow-motion / Metode pengaturan waktu sama - gerak lambat ponsel

1. Build pendulum: tie same metal nut to cotton thread, hang from door handle max two meters high, measure length from pivot to bob centre
2. Set shortest length, displace sideways with small identical angle, release without push
3. Film in slow-motion, time ten complete swings and divide by ten to get period, repeat three times and average
4. Repeat for the three longer lengths keeping bob, angle, string type and location identical
5. Record results in table and plot period squared versus length

Safety: Heights at most two meters only, use door frame with adult nearby · Household objects only: thread, tape, metal nut - no glass, no fire, no mains electricity · Clear area so bob cannot hit people; wear shoes · Adult supervision required for children

Appendix A: Claim Ledger

#	Claim	Evidence
c01	The measured gravitational acceleration is 9.62 ± 0.17 m/s^2 with confidence interval [9.30, 9.95] m/s^2.	der_04
c02	The measured oscillation period is 1.11412 ± 0.00024 s with confidence interval [1.11364, 1.11459] s.	der_01
c03	The amplitude corrected period is 1.1094 ± 0.0024 s with confidence interval [1.1048, 1.1140] s.	der_02
c04	The measured gravity differs from the reference by -1.62%.	cmp_01
c05	The bootstrap confidence interval for the period is [1.1048, 1.1140] s around 1.1094 ± 0.0024 s.	ci_01
c06	The image scale from the spoken string length is 0.461 mm/px in millimetre per pixel.	cal_02
c07	The damped sine fit explains the motion very well with low residuals.	fit_01
c08	The hypothesis test outcome is not supported.	cmp_02

Appendix B: Gate table

Gate	Status	Diagnosis	Message
G1 Track integrity hard	pass		Passed.
G2 Scale consistency soft	warn	single_scale	Only one scale method is available, so scale consistency cannot be checked.
G3 Fit quality soft	warn	smooth_model_mismatch	The fit is very close (R^2 0.9996, RMS 2.25 px), but the residuals still follow a smooth pattern (Durbin–Watson 0.08), as a wide swing or air drag causes. The uncertainty is computed with a block bootstrap, which allows for this.
G4 Physical plausibility soft	pass		Passed.
G5 Ledger completeness hard	pass		Passed.

Appendix C: Data sample

t (s)	x (m)	y (m)	valid	flag
3.054545	0.000000	0.000000	1	ok
3.363636	0.041585	0.008502	1	ok
3.672727	0.147132	0.002582	1	ok
3.981818	0.067154	0.011366	1	ok
4.290909	-0.007535	-0.001661	1	ok
4.600000	0.093643	0.010723	1	ok
4.909091	0.138227	0.004560	1	ok
5.218182	0.016459	0.005486	1	ok
5.527273	0.018082	0.004894	1	ok
5.836364	0.135720	0.005067	1	ok
6.145455	0.097713	0.010715	1	ok

t (s)	x (m)	y (m)	valid	flag
6.454545	-0.005673	-0.000540	1	ok
6.763636	0.068040	0.009987	1	ok
7.072727	0.146385	0.003459	1	ok
7.381818	0.043844	0.007164	1	ok
7.690909	0.002053	0.002036	1	ok
8.000000	0.116076	0.007368	1	ok
8.309091	0.120765	0.008279	1	ok
8.618182	0.004735	0.002805	1	ok
8.927273	0.040901	0.009103	1	ok
9.236364	0.143258	0.004180	1	ok
9.545455	0.071338	0.011708	1	ok
9.854545	-0.005024	-0.000451	1	ok
10.163636	0.091098	0.010958	1	ok
10.472727	0.136347	0.005563	0	outlier
10.781818	0.022684	0.006128	1	ok

Generated from the run's saved files. Built with Meta SAM 3.1 · Muse Spark · Muse Spark Open · Auditor: muse-spark-1.3